

Foundation (Jalapeno)

Q1. What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
 (B) $ax + bx^2 + c$
 (C) $ax^2 - bx + c$
 (D) $ax^3 + bx^2 + c$
 (E) $ax^2 + b$

Q2. Which of the following is a quadratic function in standard form?

- (A) $x^2 + 3x - 2$
 (B) $x^3 - 2x^2 + x$
 (C) $x^2 - 4$
 (D) $2x^2 + 5x$
 (E) $x^4 - 3x^2 + 1$

Q3. What is the coefficient of the x^2 term in the quadratic function $f(x) = 2x^2 + 3x - 1$?

- (A) 1
 (B) 2
 (C) 3
 (D) -1
 (E) -2

Q4. Which of the following quadratic functions has a vertex at $(0, 0)$?

- (A) $x^2 + 2x + 1$
 (B) $x^2 - 2x - 1$
 (C) x^2
 (D) $x^2 + 1$
 (E) $x^2 - 1$

Q5. What is the axis of symmetry of the quadratic function $f(x) = x^2 + 2x + 1$?

- (A) $x = -1$
 (B) $x = 0$
 (C) $x = 1$
 (D) $x = -2$
 (E) $x = 2$

Q6. Which of the following is a key feature of the quadratic function $f(x) = x^2 + 2x + 1$?

- (A) A maximum value at $(0, 0)$
 (B) A minimum value at $(-1, 0)$
 (C) A maximum value at $(-1, 0)$
 (D) A minimum value at $(0, 1)$
 (E) A maximum value at $(1, 0)$

Q7. What is the y -intercept of the quadratic function $f(x) = x^2 + 2x + 1$?

- (A) $(0, 0)$
 (B) $(0, 1)$
 (C) $(0, -1)$
 (D) $(1, 0)$
 (E) $(-1, 0)$

Q8. Which of the following quadratic functions has a y -intercept at $(0, -2)$?

- (A) $x^2 + 2x + 1$
 (B) $x^2 - 2x - 1$
 (C) $x^2 - 2$
 (D) $x^2 + 2x - 2$
 (E) $x^2 - 2x + 1$

Open Ended Questions (FRQ)**Q9.** Write the quadratic function $f(x) = x^2 + 2x + 1$ in vertex form, $f(x) = a(x - h)^2 + k$. Identify the values of a , h , and k .**Q10.** Describe the key features of the quadratic function $f(x) = x^2 + 2x + 1$, including the vertex, axis of symmetry, and y -intercept. Support your answer with calculations.

Chef's Tips

- Remind students to show all work and calculations for free-response questions.
- Encourage students to use graphing calculators or software to visualize quadratic functions.
- Emphasize the importance of identifying key features of quadratic functions, including vertex, axis of symmetry, and y -intercept.